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Speech emotion recognition model based on Bi-GRU and Focal Loss

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ABSTRACT

For the problems of inconsistent sample duration and unbalance of sample categories in the speech emotion corpus, this paper proposes a speech emotion recognition model based on Bi-GRU (Bidirection Gated Recurrent Unit) and Focal Loss. The model has been improved on the basis of learning CRNN (Convolutional Recurrent Neural Network) deeply. In CRNN, Bi-GRU is used to effectively lengthen the samples of the speech with short duration, and Focal Loss function is used to deal with the difficulties in classification caused by the imbalance of emotional categories of the samples. Through different methods for experimental comparison, weighted average recall (WAR), unweighted average recall (UAR) and confusion matrix (CM) are used as evaluation index of the algorithm. The experimental results show that the speech emotion recognition model proposed in this paper improves the recognition accuracy and the imbalance of IEMOCAP database samples, and can effectively prove that the improvement of speech emotion recognition performance is not due to the adjustment of model parameters or the change of the model topology.

1. Introduction

In the research of speech emotion recognition, a high-quality emotional corpus is a necessary prerequisite. However, in the course of practice, it's found that most emotional corpora are more or less unsatisfactory, and there are generally problems of inconsistent duration of samples and unbalanced sample categories. For example, in a typical IEMOCAP emotional corpus [1], the speech duration of the sample ranges from 0 to 30 s, and the duration span is relatively large; the proportion of happy emotion sample is small, and the proportion of neutral emotion sample is large, which has caused some inconvenience to the study of speech emotion recognition, because in machine learning or deep learning, it is often required that the number of training samples is equal and the sample size is fixed.

For the problem of inconsistent speech sample duration, the traditional solution is to extract HSFs (High Level Statistics Functions) global feature [2], such as mean and maximum value based on LLDs (Low Level Descriptors) features of speech rate, short-term zero-crossing rate, and short-term energy. Thus, a fixed-length feature vector is obtained. But this poses a problem that the timing information that can reflect the change of emotion is erased. The end-to-end neural network model for speech emotion recognition divides or fills the original speech signal to a fixed length [3], or uses a global pooling strategy [4] when the convolutional neural network outputs, and maps the variable-length input as fixed length vector. When the longer non-neutral emotion sample is divided, since the utterance contains neutral emotional words, non-neutral emotion segment, and non-emotion segment, the sub-sample after segment may not contain any emotion information or does not contain any non-neutral emotion. When the emotion recognition model learns this sub-sample, it may cause confusion of other non-neutral emotion and neutral emotion, resulting in a decrease in the accuracy of non-neutral emotion recognition [5]. When filling shorter samples, the newly filled invalid data will interfere with the parameter learning of the emotion recognition model, and the global pooling strategy will also lose some timing information [6].

The problem of unbalanced categories of speech samples is mainly due to the uneven distribution of data set categories, or extreme imbalances in some cases. Generally speaking, the sample imbalance will cause the training model to "overfit" to the emotion categories with a large number of samples, and "under-learning" to the emotion categories with a small number of samples [7], resulting in the model generalization on the test data to decrease in ability. Assuming an extreme case of unbalanced samples, the training set consists of 999 positive samples and 1 negative sample. If the training is performed without any processing directly, the classifier will directly give up the negative prediction. In this case, the clas-

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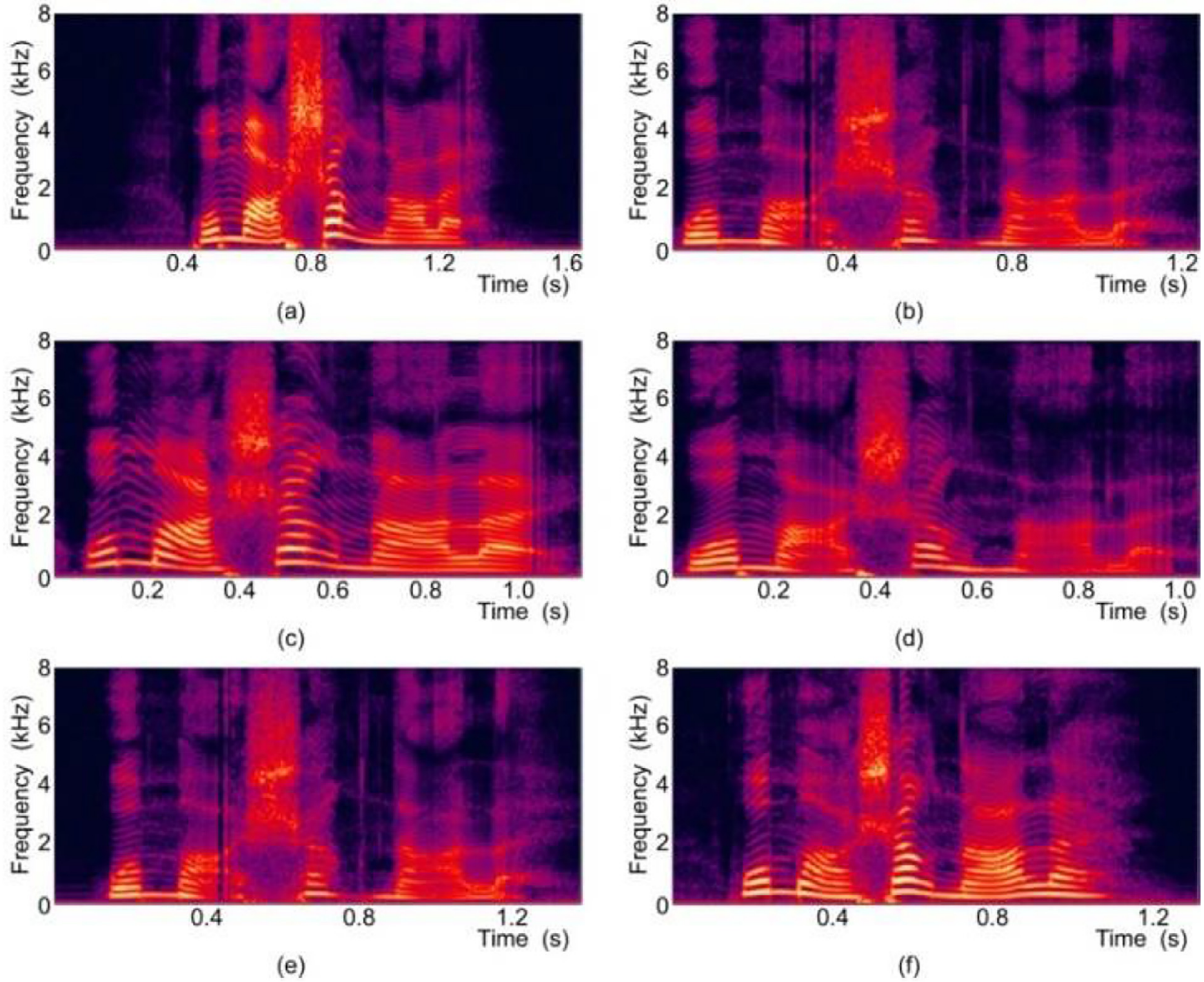

Fig. 1. Speech spectrogram of six different emotional states. (a) Anger. (b) Fear. (c) Happy. (d) Neutral. (e) Sad. (f) Surprise.

sifier can still obtain an accuracy rate of up to 99.9%, but this is meaningless [8].

For the above sample imbalance problem, this paper uses the currently popular speech spectrogram-based emotion recognition model CRNN [3,9] as the basic model, and conducts speech emotion recognition research from the aspects of inconsistent sample duration and sample category imbalance and construct a speech emotion recognition model of variable length input CRNN model based on Focal Loss.

2. Relevant theoretical bases

2.1. Speech spectrogram

The research of literature [1] and literature [6] shows that the performance of speech emotion recognition system increases with the input length within a certain length range. In speech emotion recognition, time-domain features and frequency-domain features are usually extracted for emotion recognition, but only emotion recognition from time-domain features or frequency-domain features is limited. Time-domain features do not intuitively reflect the frequency characteristics of speech signals, and frequency-domain features do not reflect changes in speech signals over time. The speech spectrogram has the advantages of both time domain features and frequency domain features. It can well represent the change of signal intensity over time in different frequency bands of speech. The horizontal axis is time and the vertical axis is frequency. The color depth indicates the energy strength of the frequency component at that time. Dark colors have high spectrogram energy, and light colors have low spectrogram energy [10].

As shown in Fig. 1, it is a speech spectrogram chart of the same text content “I'll take it immediately” in six different emotional states. It can be seen from the figure that the spectrogram chart of different emotions has obvious differences, which are embodied in formants, energy distribution, harmonic structure, and spectrogram envelope. For example, from the energy distribution in the spectrogram, the energy of anger is more fully distributed in the high-frequency part than happiness, while the energy of sadness is more concentrated in the low-frequency region than happiness, which is also very in line with our daily knowledge. When the person is in a happy mood, the voice becomes louder, and when the person is in a sad mood, the voice becomes lower. Therefore, the speech spectrogram provides theoretical basis for speech feature extraction and emotion classification.

2.2. Brief introduction of CNN and RNN networks

2.2.1. Convolutional neural networks (CNN)

CNN [11] is a biologically inspired variant of Multi-Layer Perceptron (MLP), which is proposed to solve the “gradient divergence" problem of MLP multilayer perceptron. Hubel and Wiesel discovered their unique neuronal network structure early in the study of cats' visual cortex. Their visual perception is from local to global. Each neuron actually does not perceive the global image, but perceives the local, and then the higher layer combines the local information to obtain the global information.

CNN have unique advantages in speech recognition and image processing compared to traditional neural networks due to their special structures and calculation methods, and have become one of the research hotspots in many scientific fields. The convolutional network has good translation invariance and size invariance, and it has good robustness and operation efficiency in processing applications with two-dimensional spatial characteristics, that is, local correlation characteristics [12]. In the field of image processing, the convolutional network avoids the complex pre-processing of images in traditional recognition algorithms, directly takes the original image as input. And it can extract image features such as color, texture, shape, etc. Therefore, it can completely replace traditional image processing methods and becomes one of the important methods of image feature extraction.

Generally speaking, when CNN are trained, the size of the input terminal needs to be consistent [8]. But in fact, the convolutional neural network is not sensitive to the size of the input terminal, and its training process is actually to train the convolution kernel, regardless of the size of the input terminal. The requirements for the size of the input terminal are mainly to facilitate the connection of the subsequent neural network. For example, the fully connected layer requires a fixed input size [13]. This provides a theoretical basis for the subsequent CRNN model of variable length input.

2.2.2. Recurrent neural network (RNN)

RNN [6] is a special neural network structure of cyclic connection of its own nodes, which is proposed to solve the problem that the fully connected network cannot model the time series. Most neural networks, such as traditional fully connected neural network and convolutional neural network, only establish weight connection between layers, and the nodes between each layer are disconnected, and nodes are required to be independent of each other. However, in real life, many data are related to each other. For example, the utterance of the speaker is a time series, the previous utterance and the following utterance are strongly related, and the length of the utterance is also different [14], which requires a network model that can consider the previous information and even the latter information, and can process information of any length at the same time. To solve these problems, RNNs are born.

Fig. 2 is a schematic diagram of a simple RNN structure. We see that the hierarchical structure of a RNN is simpler than that of a convolutional neural network. There is one more cycle circle than the classic neural network, which represents the data cycle in time sequence is newer, that is, the output of the neuron is used as the input of the neuron at the next moment. In essence, the RNN is to reuse a unit structure in time sequence. Therefore, the unit structure at different time series share parameters, and modeling a time series of any length will not increase the complexity of the model.

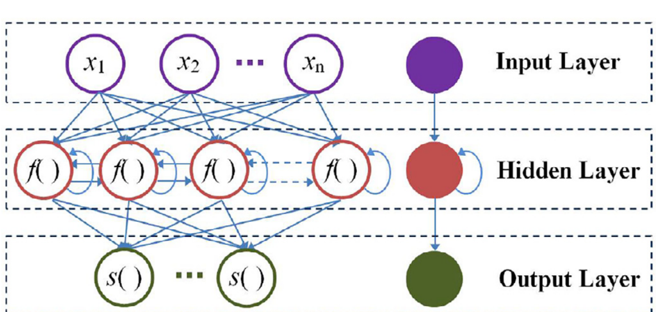

Fig. 2. RNN structure diagram.

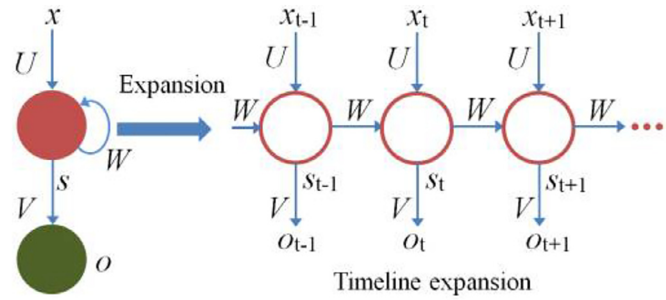

Fig. 3. Expanded diagram of the recurrent network timeline.

In order to facilitate understanding and analysis of the RNN, we will expand it according to the timeline to obtain the network structure shown in Fig. 3. In Fig. 3, *U* is the weight matrix from the input layer to the hidden layer, *V* is the weight matrix from the hidden layer to the output layer, *W* is the weight of the last value of the hidden layer as the input this time. From Fig. 3, we can see that, in addition to the output of neurons in the previous layer at time t, the input of neurons in current layer at time t in the case of forward propagation, also includes the output of the neurons in current layer at time t-1, that is, the connection between the nodes of the same layer is achieved, and the purpose of modeling the time series is achieved.

In theory, RNNs can use all previous time information to model time series of any length. But in fact, as the sequence length increases, the phenomenon of "gradient explosion" or "gradient disappearance" will appear in the RNN. To be brief, if the Sigmoid function is used as the excitation function of the neuron, the gradient of the signal with the amplitude of 1 is decayed to 0.25 of the original every time when it is transmitted backwards once. As the number of transmission increases, the gradient becomes smaller and smaller, and the network cannot receive valid signal, which is the phenomenon of "gradient disappearance". Therefore, as the time interval increases, the RNN will lose the ability to process information with a very long learning interval, that is, the "long dependence" problem of RNNs.

2.3. Bi-GRU

To solve the "long dependence" problem of traditional RNNs, Hochreiter, S. & Schmidhuber, J. [15] proposed a Long Short Term Memory (LSTM). LSTM is a special RNN for long-term information-dependence learning, which has been widely used in many sequence modeling problems and achieved great success. LSTM is different from the ordinary RNN. Compared with the single tanh activation layer in the unit structure of the ordinary RNN, its unit structure is more complicated and contains four structures that interact in a very special way.

The key to LSTM is that there is a channel to transmit the state of the cell, that is, long-term memory information, as shown in Fig. 4. There is only some simple linear interaction on this channel, which can ensure that the cell information can be completely transmitted backwards. In fact, not all LSTM cell structures are the same, and many LSTMs have made some changes to the cell structure, for example, one of the modified GRU (Gated Recurrent Unit), as shown in Fig. 5. The GRU model mixes the hidden state and the cell state, and combines the input gate and the forgotten gate into

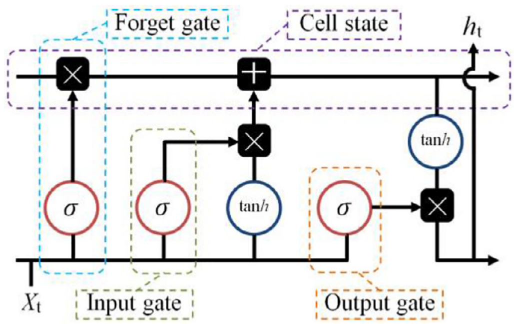

Fig. 4. LSTM structure diagram.

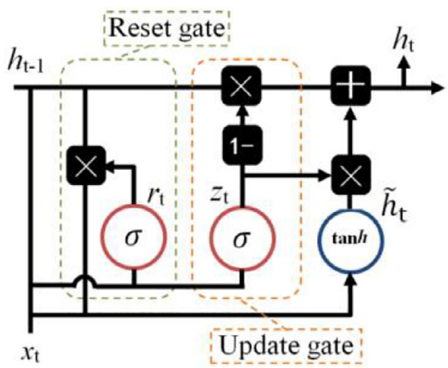

Fig. 5. GRU structure diagram.

one update gate. It is simpler than the standard LSTM model and is one of the very popular variants.

Fig. 5 can be represented with formula (1).

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \widetilde{h_t} \tag{1}$$

Among them, z_t represents the update gate, $\tilde{h}_t$ is the candidate value of the current hidden node, h_{t-1} is the activation value of the previous hidden node, and h_t is the activation value of the current hidden node. It can be seen from formula (1) that the role of the update gate is to control how much historical information h_{t-1} is "forgotten" and how much current information $\widetilde{h_t}$ is "remembered". The larger z_t is, the more current information the GRU unit can remember and the more historical information can be forgotten. The calculation of the update gate is shown in formula (2), and the candidate value of the current hidden node is calculated as shown in formula (3).

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z) \tag{2}$$

$$\widetilde{h_t} = \emptyset(W_h x_t + r_t \odot (U_h h_{t-1})) + b_h \tag{3}$$

where, x_t is the input, σ is sigmoid function, $\emptyset$ is tanh function, $\odot$ represents the multiplication of element-wise, r_t represents the reset gate. And the calculation is shown as the formula (4).

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \tag{4}$$

All of the above W and U are weight matrices used for training. The role of r_t is to control how much input information x_t and how much historical information h_{t-1} will affect $\widetilde{h_t}$. The larger r_t is, the smaller the influence of x_t on $\widetilde{h_t}$, and the greater the influence of h_{t-1}. Therefore, each GRU unit has the ability to learn long-distance information, alleviating the problem of gradient disappearance or explosion caused by traditional RNN network structure training.

This paper takes the speech spectrogram of each speech emotion category as a sequence labeling task. When faced with many speech samples of different duration, the unidirectional GRU cannot handle it well, only lengthening the samples with short speech duration simply and ignoring the information behind the speech sequence, which results in the loss of some speech features. It can be seen that the following information is very important for the same sequence. Therefore, Bi-GRU (Bidirection Gated Recurrent Unit) [16] is able to utilize both the previous information and the subsequent information, and is more suitable for processing the lengthen of sounds. It can effectively lengthen samples with a shorter phonetic duration. The structure of Bi-GRU is shown in Fig. 6.

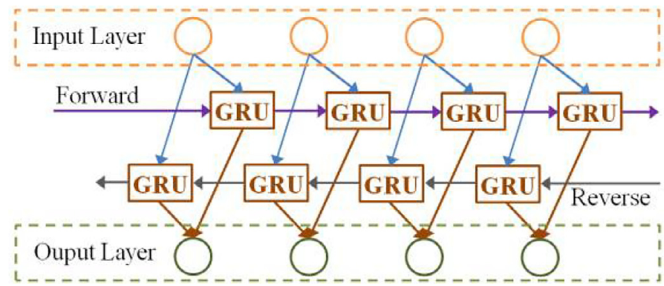

Fig. 6. Bi-GRU structure.

2.4. Focal Loss function

Focal Loss is a loss function proposed by Lin et al. [17] for the problem of low accuracy of the One-Stage detector in image object detection, which reduces the weight of a large number of negative samples in training while reducing the weight of samples that are easy to classify, making the model more focused on samples with fewer samples and difficult to classify during training. Essentially, the focus loss is a loss function to solve the classification imbalance and difficulty of classification in the classification problem.

The focus loss function is actually a modification based on the cross-entropy loss function. How was it modified? Let's first understand the cross-entropy loss function of dichotomies, as shown in formula (5) and (6).

$$CE(p, y) = \begin{cases} -\log p, & y = 1 \\ -\log(1 - p), & y = 0 \end{cases} \tag{5}$$

$$CE(p, y) = CE(p_t) = -\log p_t, \ \ p_t = \begin{cases} p, & y = 1 \\ 1 - p, & y = 0 \end{cases} \tag{6}$$

where, $y \in \{0, 1\}$ is the real label, and p is the output of the activation function. It can be seen that, for positive samples, the cross-entropy loss decreases as the output probability increases. Negative samples mean that the cross-entropy loss decreases as the output probability decreases. The cross-entropy loss function may not be optimized due to the iterative process with a large number of simple samples. So how does focus loss improve?

Control the contribution of various samples to the total loss, a control weight α_t is added, which is similar to the definition of p_t in formula (6), as shown in formula (7).

$$CE(p_t) = -\alpha_t \log p_t, \ \ \alpha_t = \begin{cases} \alpha, & y = 1 \\ 1 - \alpha, & y = 0 \end{cases}, \ \ \alpha \in (0, 1) \tag{7}$$

The contribution of positive and negative samples to the total loss can be controlled by setting the value of α. For example, if the number of samples in the $\{y = 1\}$ class is much larger than the number of samples in the $\{y = 0\}$ class, then α will take 0 to 0.5 to increase the weight of the samples in $\{y = 0\}$ class. When $\alpha = 0.5$, it is the same as standard cross entropy.

But there is a problem. It can only solve the balance problem between positive and negative samples, and cannot solve the balance problem between samples that are easy to classify and samples that are difficult to classify. Therefore, a factor γ is added on the original basis. Formula (8) is generated.

$$FL(p_t) = -(1-p_t)^{\gamma} \log p_t, \;\; \gamma \geq 0 \tag{8}$$

where, γ is called the focus factor, and $(1-p_t)^{\gamma}$ is called the modulation factor. By setting the value of γ, the contribution of misscores and easily divided samples to the total loss can be controlled. Moreover, as γ increases, the modulation coefficient will also increase.

When $\gamma > 0$, the loss of samples that are easy to classify is reduced, and the total loss is more concerned with difficult and misclassified samples. For example, if the sample of $\{y=1\}$ is misclassified as $\{y=0\}$, then $p_t = p < 0.5$, $(1-p_t)^{\gamma}$ tends to be 1, which does not affect the original loss; When the samples of this category $\{y=1\}$ are correctly classified, $p_t = p > 0.5$, the sample is easier to be classified. When p_t tends to be 1, and $(1-p_t)^{\gamma}$ tends to be 0, its contribution to the total loss is very small. When $\gamma = 0$, the focus loss is the traditional cross-entropy loss.

The focus loss is actually using a suitable function to measure the contribution of the samples that are difficult to classify and difficult to classify to the total loss. In practice, the following formulas (9) and (10) are used, that is, the forms of the above formulas (7) and (8) are integrated.

$$FL(p_t) = -\alpha_t (1-p_t)^{\gamma} \log p_t \tag{9}$$

that is:

$$L_{fl} = \begin{cases} -\alpha(1-p)^{\gamma} \log p, & y=1 \\ -(1-\alpha)p^{\gamma} \log(1-p), & y=0 \end{cases} \tag{10}$$

Through the two parameters coordinated control of α and γ, it can not only adjust the weight of positive and negative samples, but also control the weight of samples that are difficult to classify.

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2.5. Sample imbalance processing

The processing methods for sample imbalance can be divided into two aspects: "data level" and "algorithm level". These two aspects are briefly described below [18].

2.5.1. Data level

The data level processing method generally uses data sampling techniques such as random undersampling and random oversampling to make various types of samples tend to balance, that is, the number of various types of samples is basically the same. The researchers proposed a SMOTE algorithm [19] to reduce overfitting caused by oversampling through data synthesis. The oversampling no longer directly copies the samples in adding samples at this time, but artificially generate similar samples without causing overfitting of the original samples. But this method also has its shortcomings. For example, if the neighboring samples of different categories are selected for synthesis during the artificial generation process, the overlap between categories will be increased. Therefore, its improved algorithms are often used in practical application, such as Borderline-SMOTE, SMOTE based on data cleaning, etc. [20]. In addition, there are cluster-based oversampling, class-based equalization sampling, ADA-SYN, etc. [21].

2.5.2. Algorithm level

Most of the processing methods at the algorithm level are from the perspective of cost sensitivity (CS), and alleviate the problem of sample imbalance by increasing the misclassification cost of categories with fewer samples [22]. Taking the classification problem as an example, given a training set $S = (X, Y)$, there are N samples and K kinds of samples. As shown in formula (11), the CS matrix C of $K \times K$ is used to impose different penalties for different sample categories. Where, $C(y_i, x_i) = 0$, $C(y_i, y_j) \in [0, +\infty)$ means that the category y_i is wrongly classified into categories the penalty of y_j. The training goal after applying the cost becomes: training a classifier G and making the expected sum $\sum_n C(y_n, G(x_n))$ be the smallest. In addition, there are some special algorithms that usually combine the results of multiple classifiers to improve the effect of the algorithm, such as integrated algorithms.

$$C = \begin{pmatrix} C(1,1) & C(1,2) & \cdots & C(1,K) \\ C(2,1) & C(2,2) & \cdots & C(2,K) \\ \vdots & \vdots & \ddots & \vdots \\ C(K,1) & C(K,2) & \cdots & C(K,K) \end{pmatrix} \tag{11}$$

3. Speech emotion recognition model based on Bi-GRU and Focal Loss

This paper improves on the basis of the CRNN model and uses the cascade combination of CNN, Bi-GRU and Focal Loss to construct a speech emotion recognition model for sample imbalance, as shown in Fig. 7.

In the model, the same batch of speech spectrogram sequences are filled with zero value to the same length during the training phase, but the length between different batches is different, and are not filled during the prediction phase. The model uses CNN to extract the frequency domain features in the speech spectrogram, and then uses Bi-GRU to extract the time-series features in the speech spectrogram to solve the problem that the model can accept variable-length input. Finally, Focal Loss is used for network training to control the contribution of various samples and adjusted samples that are easy to classify to the total loss, and to solve the problem of imbalance of sample categories. In Fig. 7, definition $S = [x_1, x_2, \cdots, x_V, \cdots, x_T]$ is a convolutional layer input sequence, where, $S_1 = [x_1, x_2, \cdots, x_V]$is the valid part, $S_2 = [x_{V+1}, x_{V+2}, \cdots, x_T]$ is the filling part.

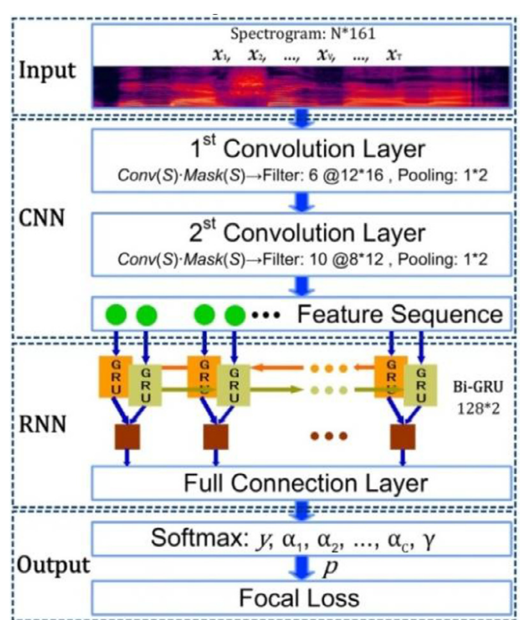

Fig. 7. CRNN model framework based on Bi-GRU and Focal Loss.

Firstly, use the element-wise multiplication between the convolutional layer output $Conv(S)$ and the masking matrix $Mask(S)$ to retain the output of S_1 and ignore the output of S_2, as shown in formula (12).

$$S_{conv} = Conv(S) \cdot Mask(S) \tag{12}$$

where, $S_{conv} = [y_1, y_2, \cdots, y_V, \cdots, y_T]$ is an output sequence equal in length to the input sequence S. The value of masking matrix $Mask(S)$ represents whether the input is valid, 1 represents the valid part, and 0 represents the filling part. In addition, in order to ensure the consistency of the input and output in the training phase with filling and the prediction phase without filling, it is necessary to pay attention to the boundary values between the effective part and the filling part to prevent these boundary values from introducing invalid information. For example, suppose S_{conv} is the input of the largest pooling layer, the pooling kernel size is $k = 2$, and the input sequence is $[y_V, y_{V+1}]$. When $y_V < 0$ and $y_{V+1} = 0$, the output will be y_{V+1} . But the correct output should be the valid value y_V, not the filling value y_{V+1}. In actual experiments, this problem will cause the neural network to fail to converge. Therefore, when $k \geq 2$, $[y_{V+1}, y_{V+k-1}]$ needs to be assigned a value of negative infinity before being input to the maximum pooling layer. With this method, the same input will produce the same output after passing through the convolutional layer and the pooling layer, regardless of whether the convolution layer input sequence S is filled.

Secondly, in the RNN, the Final-frame method [23] is used as the final output of network model. Assuming that S is the input of RNN, the final output should be the output when $t = V$. In addition, in a bidirectional RNN, the output of backward RNN should be the output at $t = 0$.

Finally, according to the length of the sentence and the proportion of samples, different inverse weights are assigned to the loss value to ensure that the length of the sentence and the proportion of samples will not cause accuracy loss to the model. The multi-class cross-entropy loss function is shown in formula (13).

$$CE = -\sum_{i=1}^{C} y_i \log p_i \tag{13}$$

In formula (13), y is the true value, p is the predicted value of the target, and C is the number of categories. With reference to the principle of two-class focus loss, the sample control weight vector $\alpha = (\alpha_1, \alpha_2, \cdots, \alpha_C)$ and the focus coefficient γ are introduced. The multi-class focus loss function is shown in formula (14).

$$FL = -\sum_{i=1}^{C} y_i \alpha_i (1 - p_i)^{\gamma} \log p_i \tag{14}$$

When p_i is the output value of Softmax, the focus loss is as shown in formula (15).

$$FL = -\alpha_1 (1 - p_i)^{\gamma} \log p_i, y_i = 1 \tag{15}$$

Given a training set S, one of the samples (X, y), then train the network by minimizing the focus loss of all (X, y) in the sample set S. The final cost function is formula (16). The optimized objective function is formula (17).

$$J(S) = \frac{1}{N} \sum_{(X,y) \in S} FL(X, y) \tag{16}$$

$$arg\ min\ J(S) \tag{17}$$

4. Experiment

4.1. Experimental design

In this paper, the IEMOCAP database is used to conduct speaker-independent 50% cross-validation experiments on six emotions: neutral, angry, happy, scared, sad, and surprised, and calculate the weighted average recall rate WAR and unweighted average recall rate UAR as the evaluation index of the algorithm. For each cross-validation, data samples from 6 sessions are used for training, and the data samples of the remaining sessions are divided into a verification set and a prediction set. Considering the imbalance of sample categories of IEMOCAP database, in order to make the emotion category with a small number of samples obtain good recognition rate, this paper aims at maximizing UAR during the training process. CNN of the model contains 2 convolutional layers and 2 maximum pooling layers. RNN is Bi-GRU with 128 × 2 nodes, which is not completely consistent with the literature [3]. In order to enhance the contrast of experiments and ensure that the improvement of the recognition effect is not caused by different network structure parameters, a “fixed length input” comparative experiment is first performed on the network model.

As mentioned above, network training is performed by minimizing the total “focus loss” of all samples. According to the proportion of the sample, sets the hyper-parameter $\alpha = (0.75,\ 0.46,\ 0.49,\ 0.36,\ 0.32,\ 0.25)$ of "focus loss" in inverse

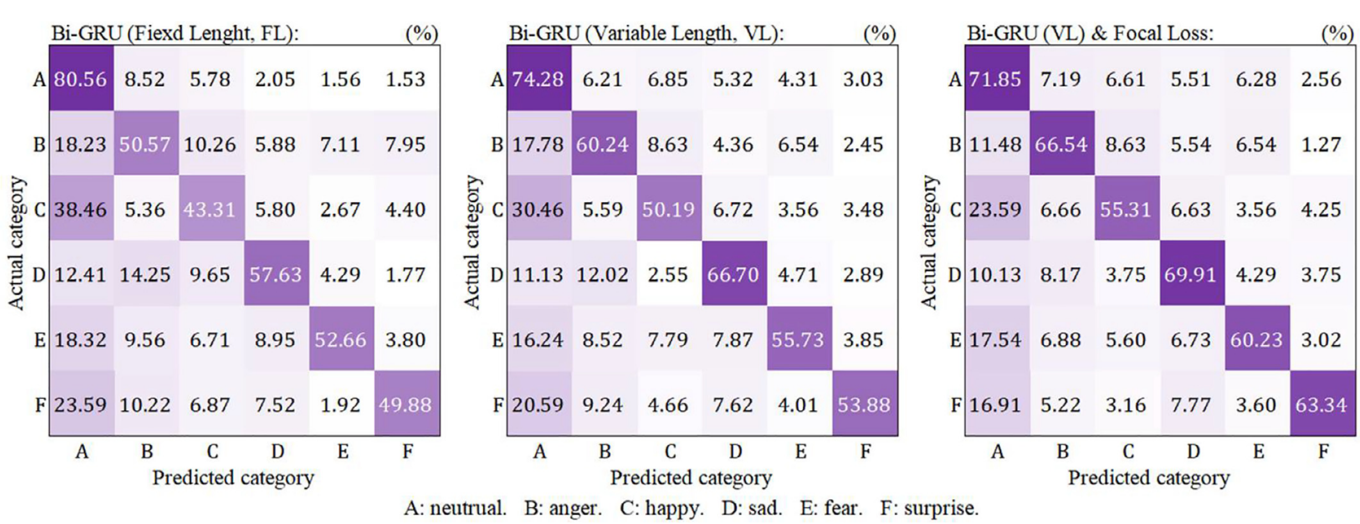

Fig. 8. Confusion matrix diagrams for different methods.

Table 1
Comparison of WAR and UAR by different methods (%).

Methods	WAR	UAR
Literature [3]	68.80	59.40
Bi-GRU(Fiexd Length, FL)	63.56	56.57
Bi-GRU(Variable Length,VL)	69.01	65.43
Bi-GRU(VL) & Focal Loss	69.73	68.75

proportion ratio, corresponding to the control weight of happiness, sadness, anger, fear, surprise, neutral emotion category respectively and the focus factor $\gamma = 2$ is set at the same time. The experimental results are shown in Table 1 and Fig. 8.

4.2. Experimental design

It can be seen from Table 1 that the “variable length input” obtains 69.01% (WAR) and 65.43% (UAR) recognition results, which is 0.21% (WAR) and 6.03% (UAR) higher than the results obtained in [3]. “Focus loss” obtains 69.73% (WAR) and 68.75% (UAR) recognition results, which is 0.72% (WAR) and 3.32% (UAR) improvement over the “variable length input” model. In addition, you can also see that “variable length input” has improved by 5.45% (WAR) and 8.86% (UAR) compared to “fixed length input”. It also shows that the improvement of system performance is not due to the adjustment of network model parameters or the change of network model topology.

Fig. 8 shows the confusion matrix of the experiment with three different methods. It can be seen that the recognition accuracy of neutral emotion has decreased, and the probability of other emotions being mistakenly recognized as neutral emotion has decreased significantly. As we analyzed earlier, because the entire speech sample is used as the input of the network, the confusion between neutral emotion and other non-neutral emotions is alleviated. These results show that “variable length input” improves the recognition accuracy of emotion recognition, and “focus loss” neural network does improve the imbalance of IEMOCAP database samples.

5. Conclusion

Aiming at the inconsistency of the duration of speech samples and the imbalance of categories of speech samples in IEMOCAP database, the speech variable length processing and speech sample category imbalance processing are studied in this paper. Speech emotion recognition model based on Bi-GRU and Focal Loss is designed. Focus loss is used for network training, which increases the contribution of small number of happiness emotion samples to the total loss of the network, and reduces the contribution of large number of neutral emotion samples to the total loss of the network, making 6 types of emotion samples obtain sufficient learning to avoid the “overfitting” problem of the emotional category with a large number of samples and the "under-learning" problem of the emotion category with a small number of samples due to the imbalanced sample category of the emotion corpus, which improves the model Recognition performance. This study provides a new way of thinking about the problems caused by the imbalance of samples in the speech emotion database.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] J.H. Tao, J. Huang, Y. Li, Z. Lian, M.Y. Niu, Semi-supervised Ladder Networks for Speech Emotion Recognition, Int. J. Autom. Comput. 16 (2019) 437–448, doi:10.1007/s11633-019-1175-x.
[2] Yatawatte Hasini, Poellabauer Christian, Sandra, Schneider, et al., Deviations of acoustic low-level descriptors in speech features of a set of triplets, one with autism, in: Conference Proceedings:. Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Annual Conference, 2018, doi:10.1109/EMBC.2018.8513289.
[3] A. Satt, S. Rozenberg, R. Hoory, Efficient emotion recognition from speech using deep learning on spectrograms, Proc. Interspeech (2017) 1089–1093, doi:10.21437/Interspeech.2017-200.
[4] H.M. Fayek, M. Lech, L. Cavedon, Towards real-time Speech Emotion Recognition using deep neural networks, in: 2015 9th International Conference on Signal Processing and Communication Systems (ICSPCS), Cairns , QLD, 2015, pp. 1–5, doi:10.1109/ICSPCS.2015.7391796.
[5] L. Jiang, P. Tan, J. Yang, X. Liu, C. Wang, Speech emotion recognition using emotion perception spectral feature, Concurr. Comput. Pract. Exp. 11 (2019) e5427, doi:10.1002/cpe.5427.
[6] H.M. Fayek, M. Lech, L. Cavedon, Evaluating deep learning architectures for Speech Emotion Recognition, Neural Networks 92 (2017) 60–68, doi:10.1016/j.neunet.2017.02.013.
[7] J. Deng, X. Xu, Z. Zhang, S. Frühholz, B. Schuller, Semisupervised autoencoders for speech emotion recognition, IEEE/ACM Trans. Audio Speech Lang. Process. 26 (1) (2017) 31–43, doi:10.1109/TASLP.2017.2759338.
[8] M. Neumann, N.T. Vu, Attentive convolutional neural network based speech emotion recognition: a study on the impact of input features, signal length, and acted speech, arXiv preprint. (2017). arXiv: 1706.00612, DOI: 10.21437/Interspeech.2017-917.
[9] W. Lim, D. Jang, T. Lee, Speech emotion recognition using convolutional and recurrent neural networks, in: 2016 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA), IEEE, 2016, pp. 1–4, doi:10.1109/apsipa.2016.7820699.
[10] Y.M.G. Costa, L.S. Oliveira, C.N. Silla, An evaluation of convolutional neural networks for music classification using spectrograms, Appl. Soft Comput. 52 (2017) 28–38, doi:10.1016/j.asoc.2016.12.024.
[11] Z.Z. Zhu, H. Chen, Y. Hu, J.S. Li, Age estimation algorithm of facial images based on multi-label sorting, EURASIP J. Image Video Process. 1 (2018) 1–10, doi:10.1186/s13640-018-0353-z.
[12] J. Zhao, X. Mao, L. Chen, Speech emotion recognition using deep 1D & 2D CNN LSTM networks, Biomed. Signal Process. Control 47 (2019) 312–323, doi:10.1016/j.bspc.2018.08.035.
[13] S. Zhang, S. Zhang, T. Huang, W. Gao, Speech emotion recognition using deep convolutional neural network and discriminant temporal pyramid matching, IEEE Trans. Multimedia 20 (6) (2017) 1576–1590, doi:10.1109/TMM.2017.2766843.
[14] S. Mirsamadi, E. Barsoum, C. Zhang, Automatic speech emotion recognition using recurrent neural networks with local attention, in: 2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2017, pp. 2227–2231, doi:10.1109/ICASSP.2017.7952552.
[15] S. Hochreiter, J. Schmidhuber, Long short-term memory, Neural Comput. 9 (8) (1997) 1735–1780, doi:10.1162/neco.1997.9.8.1735.
[16] X. Luo, W. Zhou, W. Wang, Y. Zhu, J. Deng, Attention-based relation extraction with bidirectional gated recurrent unit and highway network in the analysis of geological data, IEEE Access 6 (2017) 5705–5715, doi:10.1109/ACCESS.2017.2785229.
[17] T.Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollár, Focal loss for dense object detection, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 2980–2988, doi:10.1109/ICCV.2017.324.
[18] H. He, E.A. Garcia, Learning from imbalanced data, IEEE Trans. Knowl. Data Eng. 21 (9) (2009) 1263–1284, doi:10.1109/tkde.2008.239.
[19] N.V. Chawla, K.W. Bowyer, L.O. Hall, W.P. Kegelmeyer, SMOTE: synthetic minority over-sampling technique, J. Artif. Intell. Res. 16 (2002) 321–357, doi:10.1613/jair.953.
[20] S. Maldonado, J. López, C. Vairetti, An alternative SMOTE oversampling strategy for high-dimensional datasets, Appl. Soft Comput. 76 (2019) 380–389, doi:10.1016/j.asoc.2018.12.024.
[21] H. Ding, B. Wei, Z. Gu, Z. Yu, H. Zheng, B. Zheng, J. Li, KA-ensemble: towards imbalanced image classification ensembling under-sampling

and over-sampling, Multimed. Tools Appl. (2019) 1–18, doi:10.1007/s11042-019-07856-y.
[22] M. Simão, p. Neto, O. Gibaru, EMG-based online classification of gestures with recurrent neural networks, Pattern Recognit. Lett. 128 (2019) 45–51, doi:10.1016/j.patrec.2019.07.021.
[23] R. Szeto, X. Sun, K. Lu, J.J. Corso, A temporally-aware interpolation network for video frame inpainting, Comput. Vis.–ACCV 2019 (2018) 249–264, doi:10.1007/978-3-030-20893-6_16.

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